

Auteur: Seda Jusjajeva
Studierichting: Informatica
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$$\begin{aligned}\lim_{x \rightarrow 0} \frac{1 - \cos x + x \sin x}{2x^2} &= \frac{0}{0} \\ \stackrel{(H)}{=} \lim_{x \rightarrow 0} \frac{\sin x + x \cos x + \sin x}{4x} &= \frac{0}{0} \\ \stackrel{(H)}{=} \lim_{x \rightarrow 0} \frac{\cos x + \cos x - x \sin x + \cos x}{4} \\ &= \lim_{x \rightarrow 0} \frac{3 \cos x - x \sin x}{4} = \frac{3}{4}\end{aligned}$$

References